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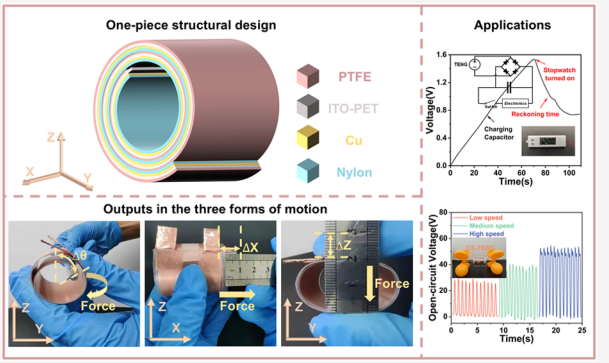
Coil-Shaped Triboelectric Nanogenerator with One-Piece Design for Multiform Energy Harvesting and Self-Powered Sensors

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ABSTRACT: Designing innovative triboelectric nanogenerators (TENGs) with simple structures, high efficiency, and multifunctionality is crucial for sustainable energy development. In this work, we introduce a coil-shaped triboelectric nanogenerator (CS-TENG) through a facile and cost-effective fabrication method. The unique one-piece design enables the CS-TENG to harvest energy from multiple motion forms, including rolling, horizontal twitching, and vertical pressing, enhancing its adaptability to complex mechanical deformations. We systematically investigate the output performance and influencing factors under different motion forms and analyze the interaction effects between these motions on the output, providing insights for the development of high-performance TENGs. The experimental results indicate that the output voltage and current in the rolling form can reach up to 142 V and 5.90 μA, respectively, with an optimum power of 0.24 mW at 20 MΩ, which is sufficient for powering small commercial electronic devices. The CS-TENG achieves an output of 142 V and 5.90 μA in rolling mode with a peak power of 0.24 mW at 20 MΩ, demonstrating its potential to power small commercial electronic devices. This work offers a versatile and highly adaptable strategy for energy harvesting and self-powered sensing applications.



INTRODUCTION

With the rapid development of multifunctional electronics (such as sensors, medical devices, and portable electronics), increasing attention has focused on the exploitation of sustainable energy harvesting devices.[1,2] Mechanical energy, abundantly available from human movement and vibrations, represents one of the most abundant resources in nature.[3,4] Typically, mechanical energy can be transformed into electricity through piezoelectric,[5,6] electromagnetic,[7,8] and triboelectric effects.[9,10] Among them, a triboelectric nanogenerator (TENG) based on Maxwell's principle of displacement currents has been demonstrated to be an effective solution for harvesting mechanical energy from the environment,[11,12] owing to the advantages of a wide choice of materials, simple structure, low cost, and admirable electricity generation.[13,14] Because of its different working modes, TENG is mainly categorized into vertical contact-separation mode,[15] horizontal sliding mode,[16] single-electrode mode,[17] and freestanding layer mode.[18] These configurations have enabled diverse applications such as self-powered sensors,[19,20] human-computer interfaces,[21] human health monitoring,[22] energy harvesting,[23] and communication devices.[24,25]

In previous reports, the energy conversion efficiency and output performance of TENGs have been significantly improved through material optimization[26,27] and innovative structural designs.[28] Among various structural designs, multilayer structures possess many advantages in terms of high flexibility, excellent shape adaptability, and enhanced energy conversion efficiency,[29,30] which are commonly regarded as one of the most promising energy harvesting approaches. For example, Xia et al.[31] fabricated a lozenge stacked TENG that was harvested from human movement. The open-circuit voltage and output power density of the MP-TENG reached 392 V and 583.75 μW/m^2, respectively. Tao et al.[32] developed a "W-tube" origami TENG with a self-recovering spring structure that was compact and highly deformable. However, such multilayer structures typically increase device volume and may induce crease formation and electrode fracture after prolonged repetitive folding, which inevitably affects the device's performance and lifespan. Ko et al.[33] designed a stackable grating-structured TENG that enhances the contact area and space utilization by converting vertical deformation into lateral sliding motion mode. The device with a 5-layer stacked structure generated an open-circuit voltage of 199.4 V and a short-circuit current of 6.56 μA. However, the multilayer stacked layers also introduce challenges, such as complex

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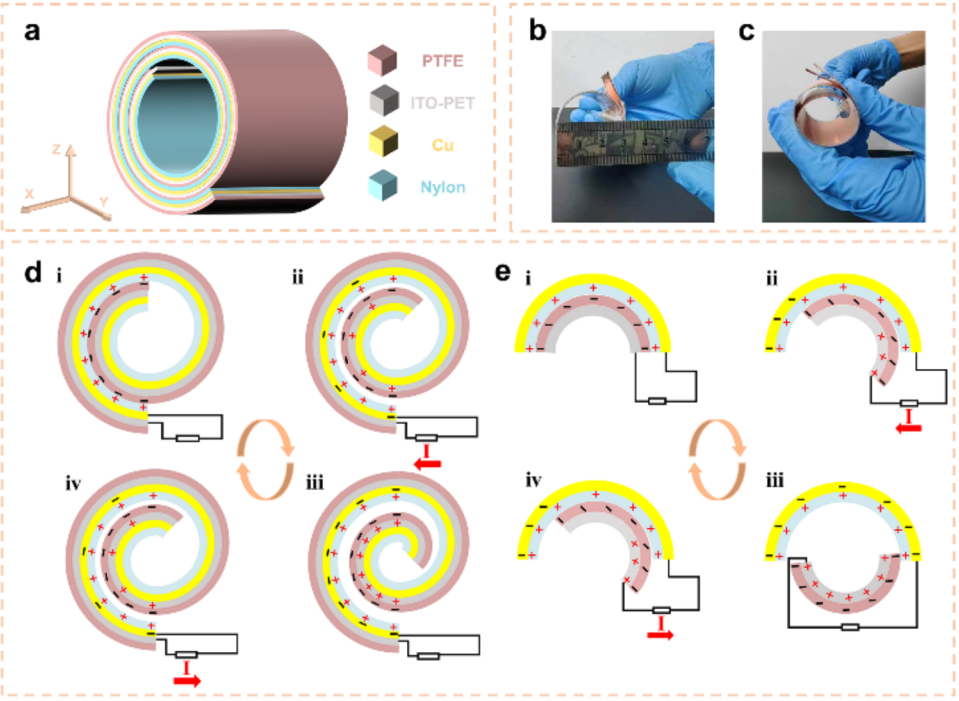


Figure 1. Structural design and working mechanism of the proposed CS-TENG. (a) Schematic diagram of the CS-TENG architecture. (b, c) Digital photographs of the CS-TENG at different rotation angles with the outer diameter remaining constant. (d, e) Working mechanism of the CS-TENG in the rolling form: working mechanism of the contact-separation mode and working mechanism of the horizontal sliding mode.

external circuit connections and reduced flexibility. Another important strategy involves designing TENGs with multiple working modes to enhance energy harvesting efficiency.[34,35] For example, Tan et al.[36] proposed an elliptical-structured TENG combining contact-separation and freestanding layer modes, while Xu et al.[37] reported a dual-mode TENG leveraging both vertical contact-separation and horizontal sliding modes. Despite these advances, most reported TENGs exhibit limited motion diversity, restricting their applicability. Therefore, it is necessary to design a new TENG with a one-piece structure, crease-free, multiple working modes, and diversified forms of motion, which can be applied to diverse real-world scenarios.

In this work, we present a coil-shaped TENG (CS-TENG). The proposed CS-TENG not only has a simple process and low production cost but, more importantly, avoids problems such as electrode breakage and life loss caused by repeated folding and pressing processes as previously reported, greatly improving the reliability and durability of the device. Moreover, it integrates two working modes (contact-separation and lateral sliding) and three motion forms (rolling, twitching, and pressing form). The output performance, influencing factors, and applications of the CS-TENG in each form of motion are systematically investigated. Furthermore, the mutual influences among the three forms of motion of CS-TENG are discussed in depth, which is significant for improving the output performance of triboelectric nanogenerators. As a proof of concept, we demonstrate a self-powered wind speed sensor based on a wind-cup-assembled CS-TENG, enabling simultaneous wind energy harvesting and speed monitoring. This research develops an innovative TENG capable of multimodal energy harvesting, significantly broadening the application of TENGs and establishing a new paradigm for developing high-performance, environmentally resilient energy harvesting systems.

EXPERIMENTAL SECTION

Fabrication of the CS-TENG. In this study, PTFE film (50 μm) and nylon film (0.05 mm) with significant differences in triboelectric series were used as triboelectric materials, and ITO-coated PET film (surface resistivity: 40 Ω sq^{-1} with a certain elasticity) was used as the supporting structure of the whole device as well as a conducive electrode. The fabrication process, illustrated in Figure S1 (Supporting Information), proceeded as follows:

First, the ITO/PET film was cut into a 200 mm × 45 mm rectangle (Figure S1(i)). Next, a 190 mm × 45 mm copper foil electrode was adhered to the PET surface (Figure S1(ii)). A nylon film of matching dimensions was then attached to the copper foil with double-sided tape (Figure S1(iii)). Subsequently, a piece of PTFE tape was pasted onto the opposite side of the ITO electrode (Figure S1(iv)). Finally, the assembled multilayer structure was rolled up into a cylindrical shape to complete the CS-TENG (Figure S1(v)). The design intentionally incorporated a 10 mm extension of the ITO/PET beyond the copper foil to facilitate the attachment of a test stick, enabling controlled rolling and twitching motions during experimental characterization (Figure S1(vi)).

Performance Measurements of CS-TENG. The output performance of the as-fabricated TENG was evaluated using three mechanical excitation methods (rolling, twitching, and pressing). The rolling and twitching motions were induced through the manual operation of the test stick. All testing was conducted under standard environmental conditions (room temperature, atmospheric pressure). A Keithley 6514 preamplifier and a Keithley 6500 digital multimeter based on the LabVIEW software platform were used to measure the open-circuit voltage and short-circuit current of the CS-TENG.

RESULTS AND DISCUSSION

Figure 1a shows a schematic diagram of the CS-TENG. Figure 1b,c presents digital photographs of the CS-TENG at different rotation angles while maintaining a constant outer diameter.

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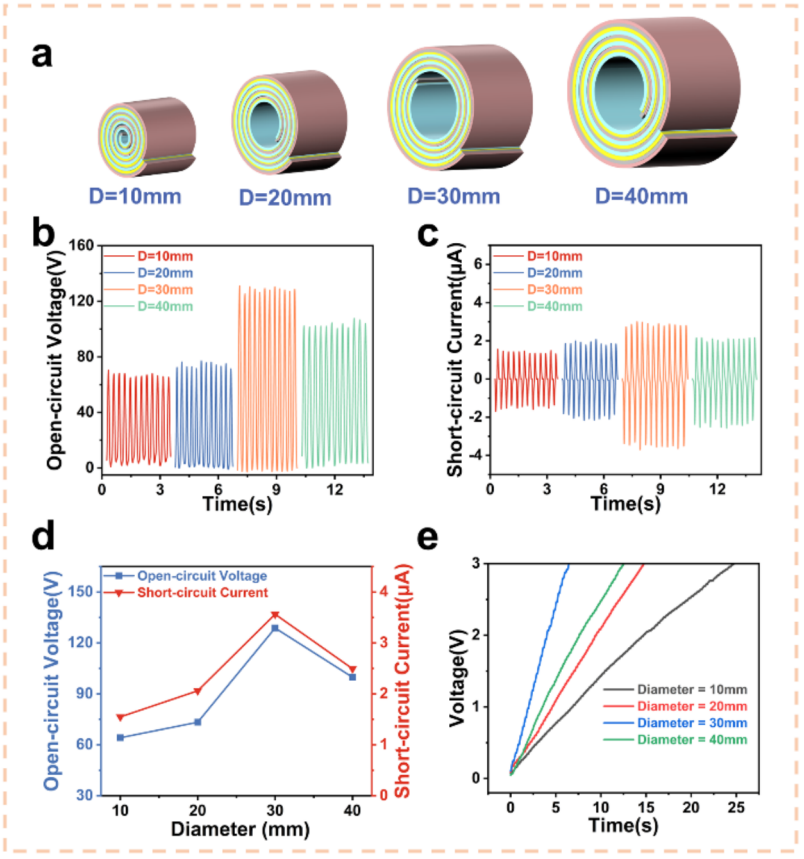


Figure 2. Effect of the diameter on the energy harvesting performance of the CS-TENG. (a) Schematic structure of the CS-TENG with different diameters. (b, c) Open-circuit voltage and short-circuit current of the CS-TENG with different diameters. (d) Open-circuit voltage and short-circuit current as functions of the diameter. (e) Charging curves of the CS-TENG with different diameters for a 3.3 μF capacitor in the rolling form.

And the outer diameter of the CS-TENG can also be adjusted from 10 to 50 mm by rolling the test stick, demonstrating excellent elasticity and environmental adaptability during the rolling-releasing process. The working principle of the CS-TENG is based on the coupling of triboelectrification and electrostatic induction.[38] Under testing conditions with a fixed outer diameter, the relative motion between the PTFE film and the nylon film will exist in two working modes of energy generation during the rolling-releasing process: the contact-separation mode and the lateral sliding mode. Figure 1d displays the working principle of the device in contact-separation mode, detailed as follows. The whole process can be divided into four consecutive states. In the initial state (Figure 1d(i)), the PTFE film and the nylon film are in close approximate contact, and surface charges will be generated between the PTFE and nylon film due to the triboelectric effect. According to the triboelectric sequence, PTFE is negatively charged, i.e., more susceptible to electrons, thereby generating positive and negative triboelectric charges on the nylon and PTFE surfaces, respectively.[39,40] Additionally, the generated triboelectric charges with opposite signs remain on the surface for a long time because of the intrinsically insulating properties of the two materials. At this time, it is in a state of charge equilibrium, so there is no current output in the external circuit. Subsequently, with an increase in the rotation angle, the separation distance between the triboelectric layers increases, which in turn breaks the charge equilibrium and drives the flow of electrons in the external circuit to balance the electrostatic field generated by the triboelectric charges (Figure 1d(ii)). Rolling continues, and a new equilibrium state is reached (Figure 1d(iii)). However, once the external force is withdrawn, the electrons return to the initial electrode as the deformation recovers (Figure 1d(iv)). Consequently, the CS-TENG is expected to produce continuous alternating current (AC) outputs under a periodic contact-separation process.

The working mechanism of the lateral sliding mode is illustrated in Figure 1e. Similarly, when the two triboelectric layers of PTFE and nylon are in contact, the contacting surfaces will be charged with equal amounts of opposite polarity due to the contact triboelectrification effect. Then, the PTFE surface will be negatively charged, while the nylon surface will be positively charged (Figure 1e(i)). Rolling the device according to the theory of electrostatic induction, electrons flow from the ITO electrode to the Cu electrode to form a potential difference to retain the original equilibrium state (Figure 1e(ii)). The rotation angle increases until a new equilibrium is reached. At this time, there is no current signal in the external circuit, and the potential difference between the electrodes reaches the maximum (Figure 1e(iii)). Subsequently, the deformation is restored, allowing the electrons to return to the original electrodes (Figure 1e(v)). It is worth noting that the contact-separation and lateral sliding modes generate currents in the same direction during the rolling process. Therefore, the generation of electrical energy is a combination of the two modes, which is significant for enhancing energy harvesting efficiency. Additionally, to gain deeper insights into the working mechanism of the CS-TENG, finite element simulations (COMSOL Multiphysics) were used

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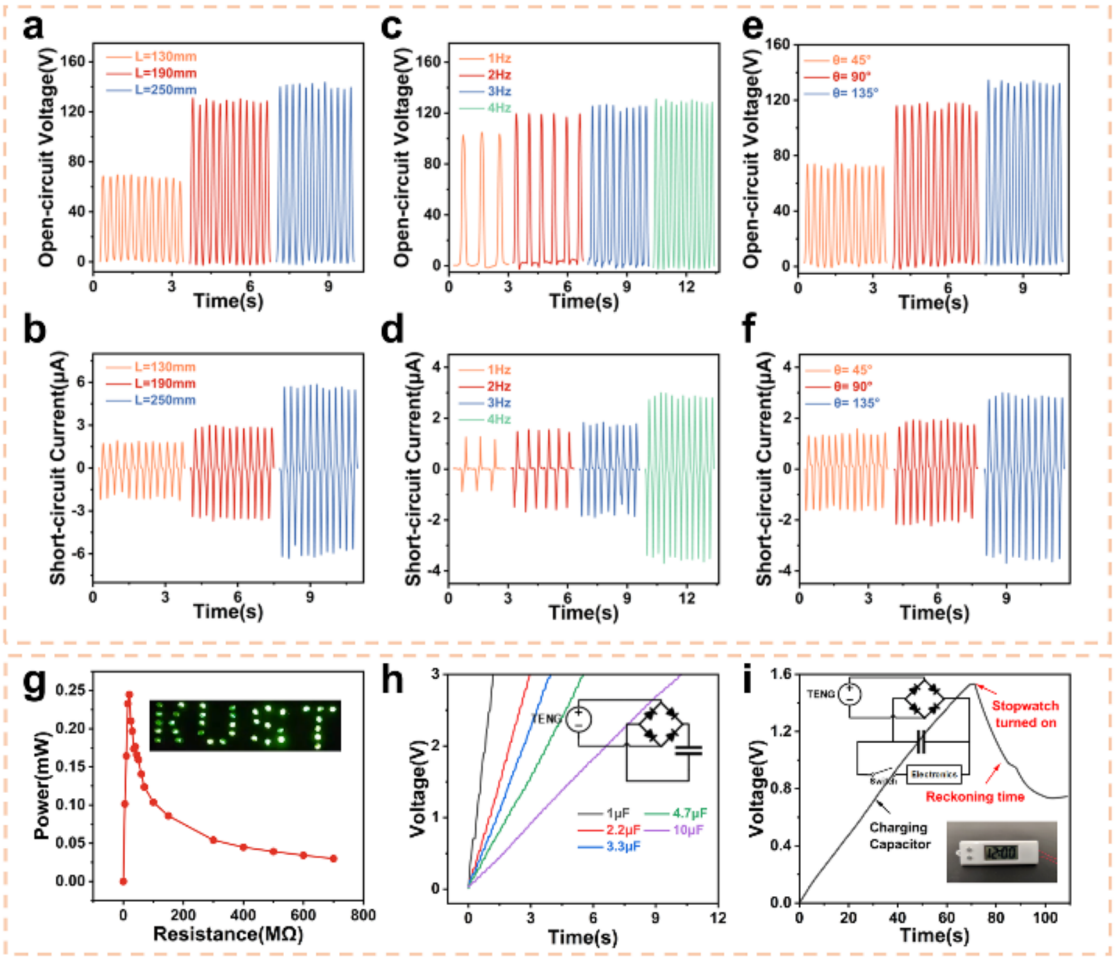


Figure 3. Output performance of the CS-TENG with an outer diameter of 30 mm in the rolling form. (a, b) Open-circuit voltage and short-circuit current of the CS-TENG with different lengths (L). (c, d) Open-circuit voltage and short-circuit current of the CS-TENG with different frequencies. (e, f) Open-circuit voltage and short-circuit current of the CS-TENG with different rotation angles (θ). Inset is the photograph showing some LEDs being powered by the CS-TENG; (g) Power output of the CS-TENG with different load resistance. (h) Charging curves of different capacitors by the CS-TENG. The inset shows the corresponding circuit diagram. (i) Charging and discharging curves of the CS-TENG for powering a commercial digital stopwatch. Inset is the photograph of the digital stopwatch.

to calculate the potential distribution of the device under two working modes. Figure S2a−h illustrates the potential difference evolution, providing direct visualization of the charge dynamics that govern electron flow in the external circuit.

The output performance of the proposed CS-TENG is intensely affected by the diameter. To reveal the variation in the output performance of the CS-TENG, we varied the outer diameter (D) of the CS-TENG from 10 to 40 mm while keeping the same length, frequency, and rotation angle of 190 mm, 4 Hz, and 135°, respectively. Schematic diagrams of the CS-TENG with different diameters are presented in Figure 2a. The open-circuit voltage and short-circuit current of the CS-TENG with different diameters are presented in Figure 2b,2c. As shown in Figure 2d, when the diameter increases from 10 to 30 mm, the open-circuit voltage increases from 64 to 128 V, and the short-circuit current increases from 1.55 to 3.56 μA. However, when the diameter increases from 30 to 40 mm, the open-circuit voltage decreases from 128 to 100 V, and the short-circuit current decreases from 3.56 to 2.50 μA. This nonmonotonic behavior arises from two competing mechanisms: (1) Larger diameters create larger interlayer gaps in the coiled state, enabling more contact surface participation in periodic contact-separation processes. As a result, there are more opportunities for charge transfer during each motion cycle, thereby achieving enhanced electrical output.[41] (2) Excessive diameter reduces interfacial contact pressure[42] and suppresses the lateral sliding working mode, ultimately limiting output performance. Then, the capability of devices with different diameters to charge a 3.3 μF capacitor was also evaluated, as presented in Figure 2e. For example, the CS-TENG with a diameter of 10 mm takes 24.6 s to charge a 3.3 μF capacitor to 3 V. In contrast, when the diameter is increased to 30 mm, the charging time is significantly shortened to only 6.5 s to reach the same voltage, which aligns well with the previous theoretical analysis.

The effects of length, frequency, and rotation angle on the output performance of the CS-TENG were systematically investigated with the outer diameter kept at 30 mm. As depicted in Figure 3a,b, the output performance of the CS-TENG with different lengths is demonstrated at a frequency of 4 Hz and a rotation angle of 135°. Obviously, with the increase in the length of the device, the open-circuit voltage rises from 61 to 142 V, and the short-circuit current increases from 2.14 to 5.90 μA. This indicates that the increase in the effective friction area of the device generates more induced charges

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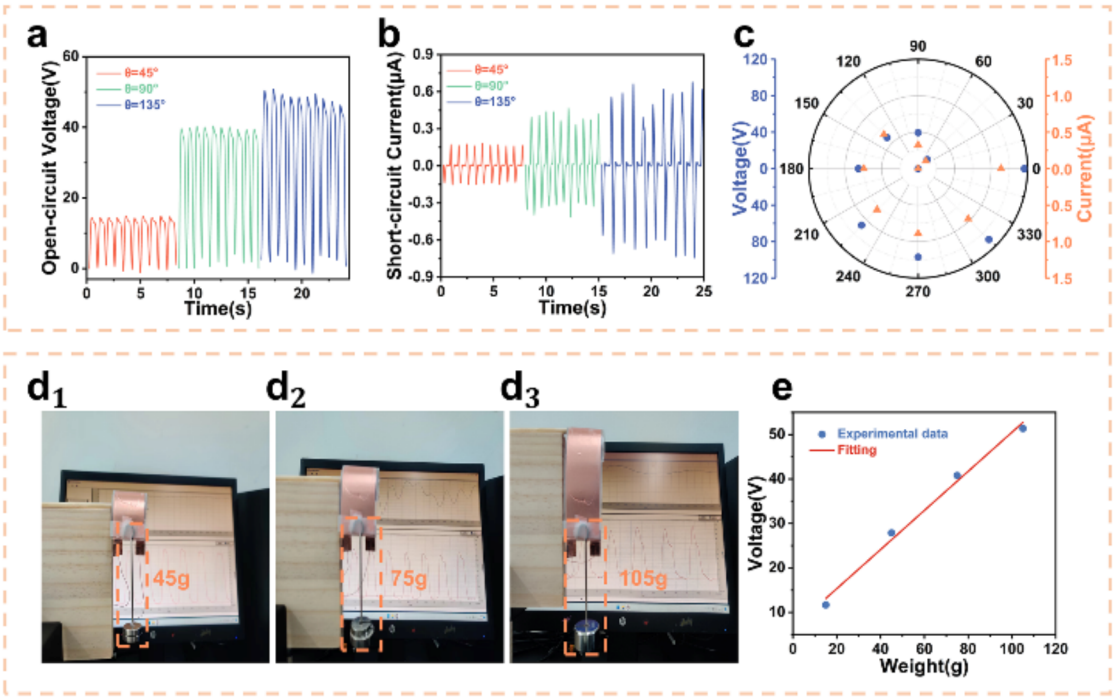


Figure 4. Output performance of the CS-TENG with the variation of the outer diameter in another rolling form. (a, b) Open-circuit voltage and short-circuit current of the CS-TENG with the rotation angles θ of 45, 90, and 135°. (c) Output performance of the CS-TENG with different rotation angles θ from 0 to 360°. (d_1−d_3) Digital photographs of CS-TENG and its output signals under different tensile forces. (e) Functional relationship between different tensile forces and output voltages.

during the rolling and release processes, resulting in improved output performance. To investigate the effect of frequency on the electrical outputs, the CS-TENG was measured with a fixed length of 190 mm and a rotational angle of 135°. As shown in Figure 3c,3d, the open-circuit voltage of the device remains consistently around 120 V. This stability can be attributed to the fact that the open-circuit voltage is determined by the triboelectric charge density and the separation distance, regardless of the influence of frequency.[43,44] In contrast, the short-circuit current depends on the amount of charge transfer per unit of time. As the frequency increases, the charge transfer becomes faster, resulting in an upward trend of the short-circuit current.[45,46] The maximum short-circuit current of 3.56 μA is achieved at 4 Hz. The three-dimensional (3D) surface plots of open-circuit voltage and short-circuit current as functions of length and frequency are plotted in Figure S3a,b, respectively. It is evident that the open-circuit voltage increases significantly with the increase in the strip length, while it remains largely unaffected by the frequency. In contrast, the short-circuit current exhibits an increasing trend with both length and frequency, which is consistent with the theoretical analysis stated above. Moreover, the magnitude of the rotation angle also affects the output performance of the CS-TENG. The rotation angle is defined here as the rotation angle (θ) of the CS-TENG in the plane formed by the Z-axis and Y-axis, as shown in Figure S4. When the device is in an unrolled state, this angle is defined as 0°. From Figure 3e,f, the open-circuit voltage and short-circuit current increase continually when increasing the rotation angle θ from 45 to 135° with the device length of 190 mm at a frequency of 4 Hz, which is mainly ascribed to the more complete and efficient contact-separation motion between the triboelectric layers with increasing angle. To better understand the effect of angle variation, the output signals were measured as the rotation angle θ varied from 0 to 360°, as shown in Figure S5. It can be observed that as the rotation angle θ increased from 0 to 135°, the open-circuit voltage and short-circuit current increased sharply, reaching peak values of 120 V and 3.56 μA, respectively. With a continuous increase in the rotation angle, the electrical outputs are basically unchanged. This is primarily due to the fact that the frictional contact area remains largely constant, leading to the saturation of the generated triboelectric charges.[47,48] To investigate the performance in harvesting mechanical energy, external load resistors were utilized to measure the output power peak (P) and output current peak (I) of the CS-TENG with the length, the diameter, the frequency, as well as a rotation angle θ of 250 mm, 30 mm, 4 Hz and 135°, respectively. As the load increases, the voltage gradually rises, while the current exhibits an opposite trend, as shown in Figure S6. The peak output power of CS-TENG was calculated as $P = I^2R$, where R is the external load resistance. As displayed in Figure 3g, the output power of the CS-TENG first increases and then decreases with the increase of the load resistance, and the output power peak reaches a maximum of 0.24 mW at an optimized load resistance of 20 MΩ. The inset demonstrates that the CS-TENG can light dozens of LEDs (Video S1, Supporting Information). Figure 3h shows charging curves of a few capacitors with different capacitances powered by the CS-TENG, and the inset shows the circuit diagram of capacitor charging. The TENG charges a 10 μF capacitor to 3 V in 10 s, while it takes about 1.2 s to charge a 1 μF capacitor to the same voltage of 3 V. The results illustrate that the smaller the capacitor, the higher the voltage it can be charged to at the same time. Figure 3i and Video S2 display that charging a 47 μF capacitor to 1.5 V in 70 s can power a commercial stopwatch for 15 s.

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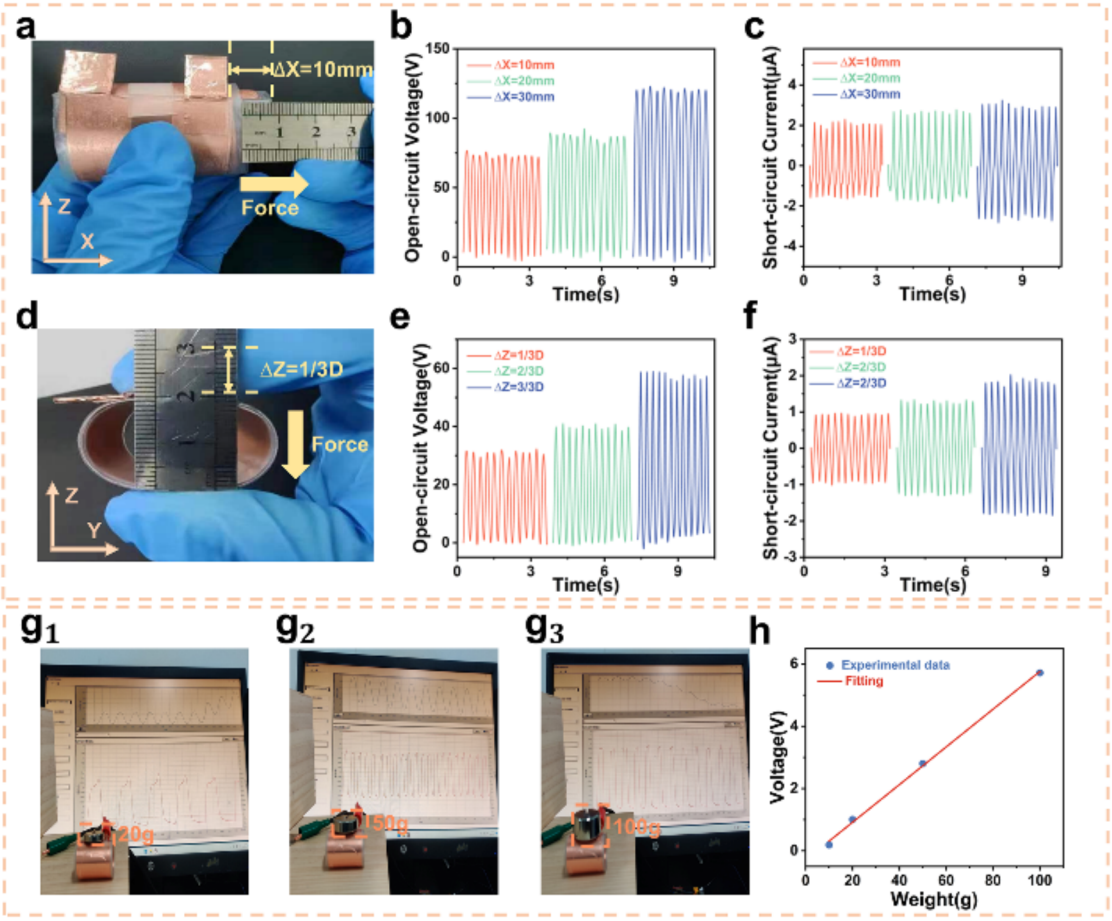


Figure 5. Output performance of CS-TENG in the horizontal twitching and vertical pressing forms. (a) Photograph of the CS-TENG with a twitching displacement of 10 mm. (b, c) Open-circuit voltage and short-circuit current of the CS-TENG with different twitching displacements. (d) Photograph of the CS-TENG with a pressing displacement of 1/3D. (e, f) Open-circuit voltage and short-circuit current of the CS-TENG with different pressing displacements. (g_1–g_3) Digital photographs of the CS-TENG and its corresponding output signals at different weights. (h) Different weights are given as a function of output voltage.

To expand the application range of the CS-TENG, the output performance of another rolling form with a variable outer diameter was also investigated. In this case, the outer diameter of the device decreases with the increase in the rotation angle. Figure 4a,4b displays the open-circuit voltage and short-circuit current of the CS-TENG with a length of 190 mm at an excitation frequency of 1 Hz under different rotation angles θ. It can be seen that both the open-circuit voltage and short-circuit current increase from 14 to 47 V and from 0.16 to 0.66 μA when increasing the rotation angle θ from 45 to 135°. Unlikely the previously described constant-diameter rolling form, this configuration primarily relies on the lateral sliding operation mechanism during this rolling process (Figure S7). The effective friction area increases with the increase of the rotation angle, and therefore, the electrical output is higher. Additionally, the output performance of the CS-TENG was tested by changing the rotation angle from 0 to 360°, and it is noticeable that the open-circuit voltage and short-circuit current show an increasing trend as the angle increases (Figure 4c). The high sensitivity of the output signals to the angle endows its potential not only in energy harvesting but also in self-powered sensing. For example, the CS-TENG could be used as a self-powered tensile sensor, which might convert the change in the rotation angle into a change in the tensile force within a specific angle range. Figure 4d_1–d_3 shows digital photographs of the CS-TENG with a length of 250 mm under different tensile forces and its corresponding output signals. As shown in Figure 4e, the output voltage of the CS-TENG increases with the increase of the tensile force in the range of 15–105 g, and there is a well-linear relationship between the tensile force and the voltage, and the tensile force sensitivity $\Delta V/\Delta g$ reaches 0.44 Vg^{-1}.

In the practical environment, owing to the complexity and variability of environmental conditions, the device may be subjected to horizontal or vertical excitation. To comprehensively evaluate the performance of the CS-TENG under such conditions, we systematically investigated its output characteristics under these two excitation conditions in both horizontal twitching and vertical pressing forms. The physical diagram of a device with a length (L), diameter (D), and horizontal twitching displacement (Δx) of 190, 30, and 10 mm, respectively, is illustrated in Figure 5a. As the horizontal twitching force increases, the electrical outputs of the device exhibit an increasing trend, and the maximum electrical output is obtained at a twitching displacement of 30 mm and a frequency of 4 Hz, with an open-circuit voltage and a short-circuit current of approximately 120 V and 2.93 μA, respectively (Figure 5b,5c). A digital photograph of the device with a vertical pressing displacement (Δz) of 1/3D is shown in Figure 5d. During the pressing and releasing processes, the

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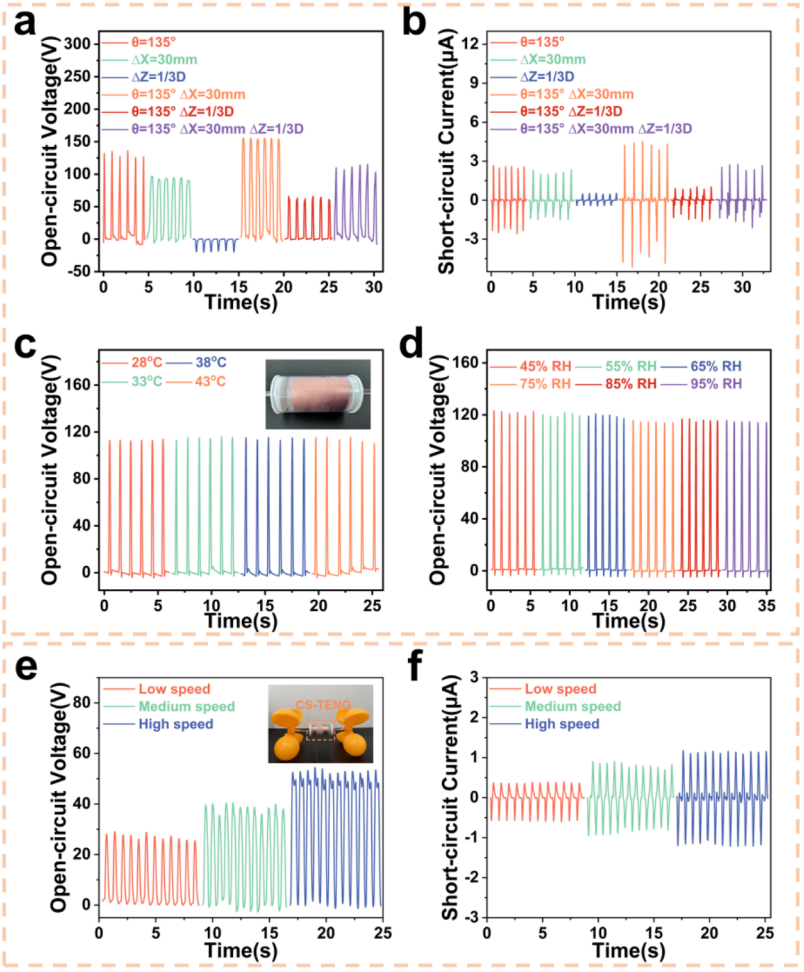


Figure 6. Output performance of CS-TENG in multiple hybrid forms. (a) Open-circuit voltage of the CS-TENG under six forms of motion. (b) Short-circuit current of CS-TENG of CS-TENG under six forms of motion. (c) Open-circuit voltage of the encapsulated device under varying temperature conditions. (d) Open-circuit voltage of the encapsulated device under varying humidity conditions. (e, f) Open-circuit voltage and short-circuit current of the CS-TENG under different wind speeds.

device exhibits excellent elastic recovery capability, which facilitates sustained and effective electrical output in real-world applications. Figure 5e,5f depicts the open-circuit voltage and short-circuit current of the device at different pressing displacements at a fixed frequency of 4 Hz. The electrical output of the device increases as the pressing displacement increases. When the pressing displacement reaches its full diameter, the open-circuit voltage is approximately 55 V, and the short-circuit current is around 1.8 μA. It can be seen that the open-circuit voltage of the CS-TENG shows an analogous linear relationship within a certain pressure range, suggesting its potential as a self-powered weight sensor. As shown in Figure 5g_1−g_3, the open-circuit voltage of the CS-TENG (with a length of 250 mm) is plotted at different weights. As shown in Figure 5h, as the weight is changed from 10 to 100 g, the voltage and weight of the CS-TENG show an excellent linear relationship with a weight sensitivity $\Delta V/\Delta g$ of 0.06 Vg^{-1}.

All of the previous experimental analyses were based on the assumption that the CS-TENGs are subjected to a single form of motion (either rolling, twitching, or pressing form) without consideration of their interaction. Considering that CS-TENG may be subjected to multidirectional excitation in practical application environments, we further investigated the energy harvesting performance of the device under combined motion modes as shown in Figure 6a,6b. Under the combined effect of twitching (displacement of 30 mm) and rolling (angle of 135°) motions, the CS-TENG reaches a maximum open-circuit voltage of approximately 155 V and a short-circuit current of 4.38 μA. It can be observed that the synergy of these two forms of motion significantly enhances the energy output. This can be attributed to the fact that the output of the CS-TENG is a combination of the two working modes (contact-separation and horizontal sliding), as illustrated in Figure 1d,1e. However, rolling motion causes an increase in the gap between the friction layers, preventing some friction surfaces from achieving effective relative sliding. Therefore, although the electrical output of the CS-TENG is enhanced, it does not reach the superposition value of the outputs from the two single modes. However, when the device is subjected to pressing (displacement of 1/3D) and rolling (angle of 135°) motions, the open-circuit voltage and short-circuit current reach approximately 64 V and 1.35 μA, respectively. Obviously, the output performance is significantly degraded when the device is subjected to a pressing action. This suppression effectively stems from two factors: (1) Charge cancellation - the pressing motion reduces the gap between the friction layers, while the rolling motion increases the gap, leading to partial neutralization of accumulated charges. (2) Structural deformation - the pressing

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motion induces uneven contact pressure and reduces the contact area between the friction layers during rolling motion, further limiting charge transfer efficiency. Consequently, the electrical output under combined pressing-rolling conditions deviates from a simple linear subtraction of the outputs from the two individual modes. To further validate the reliability of our observations, three additional independent experiments were conducted to investigate the device's energy harvesting performance under combined motion conditions (see Supporting Figure S8). The results demonstrate excellent consistency with Figure 6a,6b, confirming that these interaction phenomena represent intrinsic characteristics of the CS-TENG rather than experimental artifacts.

Therefore, to maximize the energy harvesting efficiency and operational reliability in practical applications, we designed a constrained cylindrical shell structure (30 mm in diameter × 60 mm in length) to encapsulate the as-fabricated CS-TENG device (inset of Figure 6c). This innovative encapsulation design serves three critical functions: (1) preventing stress in the vertical direction, effectively mitigating charge cancellation effects; and (2) providing effective environmental shielding and maintaining stable electrical output under varying ambient conditions. As shown in Figure 6c, the encapsulated device exhibited remarkable thermal stability, maintaining a consistent open-circuit voltage of approximately 116 V (fluctuation <4%) when the temperature rose from 28 to 43 °C. Similarly, the open-circuit voltage remained stable at approximately 125 V (fluctuation <5%) as the relative humidity increased from 45 to 95% RH, as depicted in Figure 6d. These results demonstrate that the constrained shell design significantly enhances the device's environmental tolerance and improves practical reliability. In contrast, the unencapsulated device suffered significant performance degradation: a 43% voltage reduction (123−70 V) under temperature variations (Figure S9a) and a 44% decrease (125−70 V) with increasing humidity (Figure S9b).

Based on the environmental adaptability optimizations of the device, we integrated the as-fabricated CS-TENG with a confining shell and a wind cup, as illustrated in the inset of Figure 6e. When the wind cups are subjected to wind force in a specific direction and at a certain angle, the friction layers undergo continuous contact-separation and relative sliding under the combined effects of rolling and twitching modes without considering the charge cancellation caused by pressing motion as well as environmental variations (Video S3, Supporting Information). As the wind speed increases, the open-circuit voltage increases from 28 to 53 V, and the short-circuit current follows the same trend from 0.58 to 1.12 μA, as displayed in Figure 6e,f. Notably, the open-circuit voltage and short-circuit current outputs exhibit an excellent linear relationship with wind speed (Figure S10), indicating that the fabricated CS-TENG can be used as a self-powered airspeed sensor for environmental meteorological monitoring.

CONCLUSIONS

In summary, we have developed a coil-shaped triboelectric nanogenerator featuring a one-piece structural design that requires neither complex fabrication processes nor external circuit connections. Furthermore, the as-fabricated CS-TENG possesses two working modes (contact-separation mode and horizontal sliding mode) with three distinct motions (rolling, twitching, and pressing), which can be adapted to complex mechanical deformation and improve the adaptability and flexibility of environmental energy harvesting. The experimental results indicate that the output voltage and current in the rolling form can reach up to 142 V and 5.90 μA, respectively, with an optimum power of 0.24 mW at 20 MΩ, sufficient for powering small commercial electronic devices. Additionally, the interactions among the three forms of motion were explored, revealing that the signal generated by pressing action results in charge cancellation of those produced by rolling and twitching actions, providing insights into the development of high-performance power generation and self-powered sensing systems. Finally, we assemble the wind cup with the as-fabricated CS-TENG, demonstrating its potential as an energy harvesting device and for environmental air detection or monitoring. This work introduces a novel-structured TENG, which not only advances the development of energy technologies but also provides critical technological support for the energy supply of micronano electronic devices and the practical application of intelligent monitoring in the era of the Internet of Things (IoT).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.langmuir.5c01489.

Dozens of LEDs powered by CS-TENG (Video V1) (MP4)

Commercial stopwatch powered by CS-TENG (Video V2) (MP4)

Self-powered wind speed sensing application of CS-TENG (Video V3) (MP4)

Preparation process of the proposed CS-TENG (Figure S1); potential distribution of the CS-TENG in contact-separation mode and horizontal sliding mode (Figure S2); open-circuit voltage and short-circuit current as a function of length and frequency (Figure S3); definition the rotation angle (θ) of the CS-TENG (Figure S4); output performance of the CS-TENG with a rotation angle from 0 to 360° at a constant outer diameter of 30 mm (Figure S5); output voltage and output current of the CS-TENG at different loads (Figure S6); operating mechanism of the CS-TENG in rolling form with variable outer diameter (Figure S7); output performance of the CS-TENG under six forms of motion through four consecutive cyclic test sets (Figure S8); open-circuit voltage of the unencapsulated device under varying temperature conditions and varying humidity conditions (Figure S9); and open-circuit voltage and short-circuit current as a function of different wind speed levels (Figure S10) (PDF)

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Notes

The authors declare no competing financial interest.

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